

Grade 7
Unit 9
Lesson 5 Practice

Name Answer Key
Date _____
Period _____

Evaluate the expression.

1. $10.3 + (0.6 - 2.8)$

$$10.3 - 2.2$$

$$8.1$$

2. $|(-4.25 \times 9) + 8|$

$$|-38.25 + 8|$$

$$|-30.25|$$

$$30.25$$

3. $-\frac{9}{5} \times 40 - \frac{3}{8} \times 40$

$$\left(-\frac{9}{5} \times \frac{40}{1}\right) - \left(\frac{3}{8} \times \frac{40}{1}\right)$$

$$\left(-\frac{360}{5}\right) - \left(\frac{120}{8}\right)$$

$$-72 - 15$$

$$-87$$

4. $\left(5\frac{1}{2}\right)(8^2) \times (-1^2)$

$$5.5(64) \cdot -1$$

$$352(-1)$$

$$-352$$

5. $-7\frac{2}{3} \times 6.9 \div (-0.2)$

$$-\frac{23}{3} \times \frac{6.9}{1} \times -\frac{1}{0.2}$$

$$\frac{158.7}{0.6}$$

$$264.5$$

Using properties of operations, determine whether the pairs of expressions are equivalent.

6. $-\left(\frac{-9}{-54}\right) - 1 \times 6$

$$-\frac{9}{54} - (1 \times 6) \rightarrow -\frac{9}{54} - \frac{6}{1} \rightarrow -\frac{9}{54} - \frac{324}{54}$$

$$\rightarrow -\frac{333}{54} \rightarrow -\frac{37}{6} = -6\frac{1}{6}$$

equivalent

$$-\left(\frac{-9}{-54}\right) - 6 \times 1$$

$$-\frac{9}{54} - (6 \times 1) \rightarrow -\frac{9}{54} - \frac{6}{1} \rightarrow -\frac{9}{54} - \frac{324}{54}$$

$$\rightarrow -\frac{333}{54} \rightarrow -\frac{37}{6} = -6\frac{1}{6}$$

7. $\frac{5\left(\frac{3}{4} + \frac{13}{16}\right)}{4}$

$$\frac{5\left(\frac{12}{16} + \frac{13}{16}\right)}{4} \rightarrow \frac{5 \times \frac{25}{16}}{4} \rightarrow \frac{125}{16} \times \frac{1}{4} = \frac{125}{64} = 1\frac{61}{64}$$

not equivalent

$$\frac{5\left(\frac{3}{4}\right) + \frac{13}{16}}{4}$$

$$\frac{5}{1} \left(\frac{3}{4}\right) + \frac{13}{16} \rightarrow \frac{15}{4} + \frac{13}{16} \rightarrow \frac{60}{16} + \frac{13}{16}$$

$$\frac{73}{16} \times \frac{1}{4} = \frac{73}{64} = 1\frac{9}{64}$$

8. $(5.9 - 6) + (4.9 + 8) - (-1.9 - 4)$

$$-0.1 + 12.9 + 5.9 = 18.7$$

equivalent

$$(4.9 + 8) + (5.9 - 6) - (-1.9 - 4)$$

$$12.9 - 0.1 + 5.9 = 18.7$$

9. $\left(2.9 + \frac{2}{9}\right) + \left(-4.7 + \frac{7}{9}\right)$

$$\left(2\frac{9}{10} + \frac{2}{9}\right) + \left(-4\frac{7}{10} + \frac{7}{9}\right)$$

$$\left(2\frac{81}{90} + \frac{20}{90}\right) + \left(-4\frac{63}{90} + \frac{70}{90}\right)$$

$$3\frac{11}{90} - 3\frac{83}{90} = -\frac{72}{90} = -0.8$$

$$(2.9 - 4.7) + \left(\frac{2}{9} + \frac{7}{9}\right)$$

equivalent

$$-1.8 + 1 = -0.8$$

10. $-\left|\frac{3}{5} - 5\frac{1}{3}\right|$ $-\left|\frac{3}{5} - \frac{16}{3}\right| \rightarrow -\left|\frac{9}{15} - \frac{80}{15}\right| \rightarrow -\left|-\frac{71}{15}\right| = -\frac{71}{15}$

$$-\left| -5\frac{1}{3} + \frac{3}{5} \right|$$


same expression as above

equivalent